

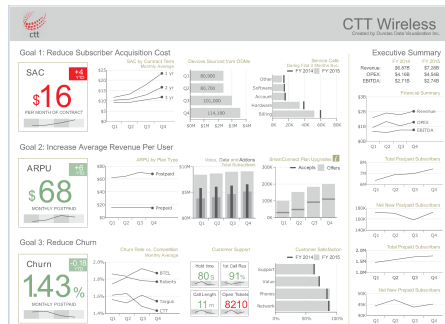
Chapter 27 Case Study

Telecom Operator Executive Dashboard

Dashboard design for executive decision-making

Presenter: Tran Nguyen My Anh
Course: Data Visualization / Dashboard Design
Source: *The Big Book of Dashboards*, Chapter 27

The structure follows the business questions: SAC, ARPU, churn, and overall health.



Scenario and Business Context

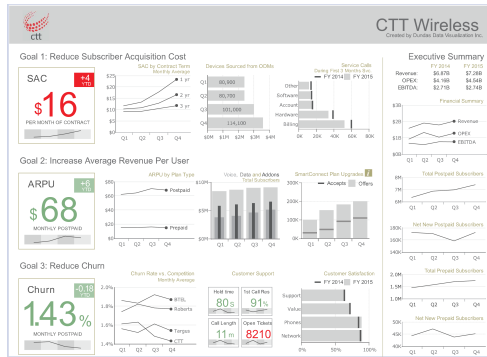
Who uses it?

- Executives and senior management at a telecom company.
- Used in management, board, or stakeholder meetings.
- The goal is executive-level monitoring, not deep interactive analysis.

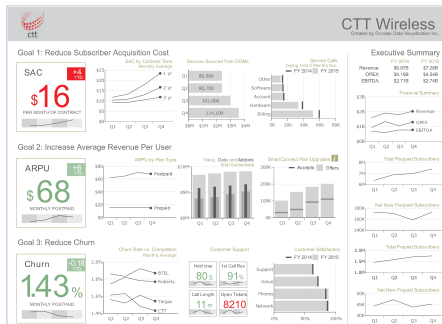
What must it answer quickly?

- Is the company on track against strategic goals?
- Which metrics are improving or worsening?

Use this as a monitoring view: status first, diagnosis second.



Dashboard Design Logic

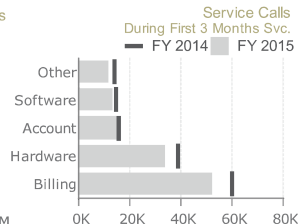
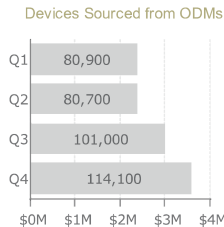
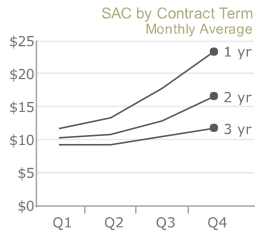


The dashboard is structured around four information zones

- Row 1: **Reduce Subscriber Acquisition Cost.**
- Row 2: **Increase Average Revenue Per User.**
- Row 3: **Reduce Churn.**
- Right column: **Executive Summary** for finance and subscriber status.

Goal 1: Reduce SAC

Goal 1: Reduce Subscriber Acquisition Cost



Status

- **\$16**/month of contract.
- **+4 YTD** above benchmark.

Drivers

- Contract term changes cost.
- ODM device mix adds pressure.

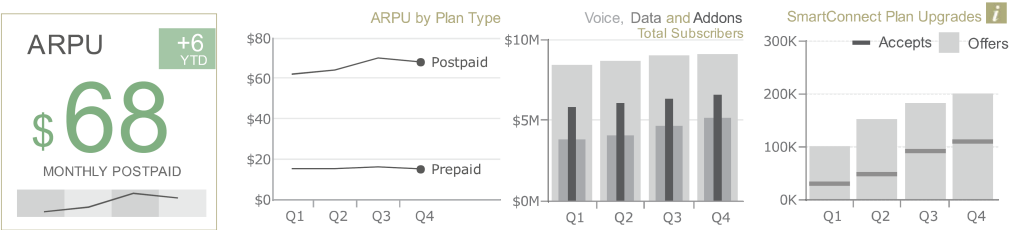
Operational signal

- Service calls show burden.
- Target lines show FY2014 gap.

Priority: identify which acquisition driver is pushing SAC above the benchmark.

Goal 2: Increase ARPU

Goal 2: Increase Average Revenue Per User



Status

- \$68 monthly postpaid.
- +6 YTD revenue lift.

Revenue mix

- Postpaid/prepaid split.
- Add-ons show contribution.

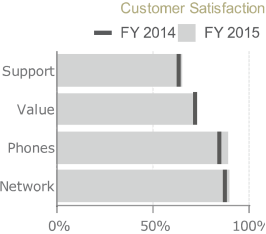
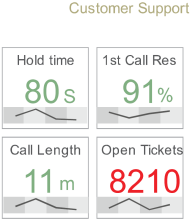
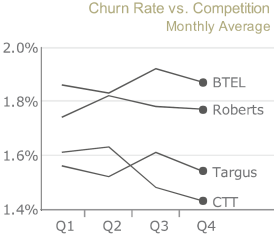
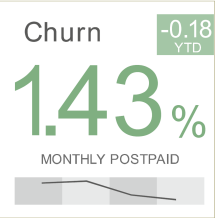
Conversion

- Offers compared with accepts.
- Gaps show upgrade room.

| Read ARPU growth through mix, add-on contribution, and upgrade acceptance.

Goal 3: Reduce Churn

Goal 3: Reduce Churn



Status

- 1.43% monthly churn.
- -0.18 YTD improvement.

Competitive view

- CTT benchmarked vs peers.
- Direct labels avoid a legend.

Retention risk

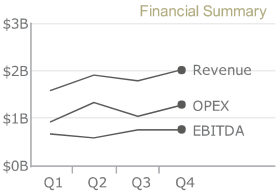
- Support KPIs show friction.
- Satisfaction flags churn risk.

The churn headline is favorable; support and satisfaction explain whether it is durable.

Executive Summary

Executive Summary

	FY 2014	FY 2015
Revenue:	\$6.87B	\$7.28B
OPEX:	\$4.16B	\$4.54B
EBITDA:	\$2.71B	\$2.74B



Finance: revenue, OPEX, and EBITDA stay visible as executive context.



Executive read

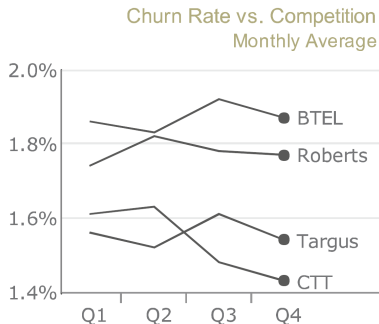
- 01 Financial health
- 02 Subscriber base
- 03 Net additions

Use: company context for the three goal rows.

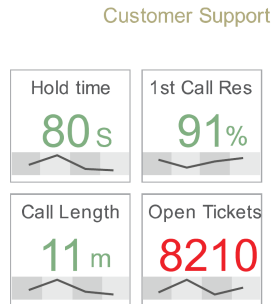
Do not use: detailed diagnosis by itself.

The summary column gives context; the goal rows explain what to act on.

Why This Works: Compact but Informative



Direct labels. Line-end labels remove the legend lookup and keep each series attached to its endpoint.



Small KPI cards. Sparklines add trend context while keeping four support measures in a tight space.

Compact visuals work when labels, trends, and comparisons are resolved inside the chart.

Layout Supports Quick Reading

Design principles used

- **Proximity:** related charts are placed near one another.
- **Closure:** each goal is perceived as a separate group.
- **Alignment:** rows and columns make scanning predictable.
- **Visual hierarchy:** KPI cards appear first, summary stays on the right.

Reading paths: scan horizontally to analyze one goal; scan the right column for company-wide status.

The layout supports two reading paths: across each goal row, then down the summary column.

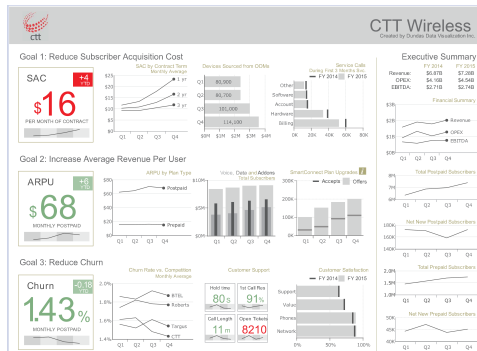
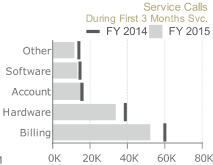
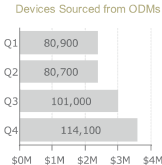
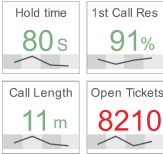


Chart Type Selection

Goal 1: Reduce Subscriber Acquisition Cost



Customer Support



Goal row: KPI card, trend line, comparison bars, and target line are read as one decision unit.

Line charts

Trends over time.

Bar charts

Category and target comparisons.

Sparklines

Compact trend cues.

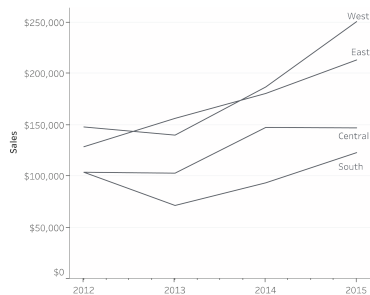
Target lines

Current values versus benchmark.

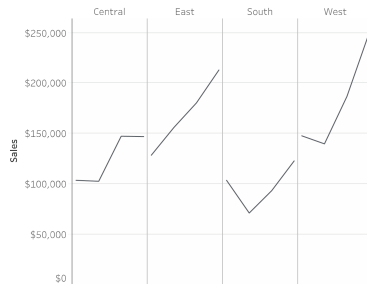
Chart type follows the question: trend, comparison, compact status, or benchmark.

Critique and Design Lessons

Sales by Region



Sales by Region, 2012-2015



Problem: overlapping lines are hard to trace when several regions cross in the same plot.

Other critiques: red/green needs a non-color cue, and emphasis colors should be used sparingly.

| Accessibility and line separation are design requirements, not cosmetic refinements.

Fix: split dense series into small multiples so each region keeps a consistent time axis.

Conclusion

What Chapter 27 teaches

- A good dashboard helps decision-makers understand what is improving, what is worsening, and where action is needed.
- This dashboard succeeds because it follows three clear business goals.
- The three-row plus one-summary-column layout makes scanning fast.
- Small charts, direct labels, and limited color reduce clutter.
- The design still requires care with accessibility, overlapping lines, and chart consistency.

| The goal is not more data on screen; it is faster decisions from the right evidence.

References

- Steve Wexler, Jeffrey Shaffer, Andy Cotgreave. *The Big Book of Dashboards*, Chapter 27: Telecom Operator Executive Dashboard.
- Dundas Data Visualization sample dashboard: Telecom / Wireless Executive Strategy Dashboard.
- O'Reilly online excerpt for Chapter 27:
<https://www.oreilly.com/library/view/the-big-book/9781119282716/c27.xhtml>
- Dashboard design references: related metric grouping, proximity, clutter reduction, clear KPI structure, and accessibility.

Thank you